

29**D**

When speed of electrons is halved, the KE of electrons is reduced to $\frac{1}{4}$ of its original value.

Hence, maximum photon energy will be reduced to $\frac{1}{4}$ of its original value as well.

$$E'_p = \frac{hc}{4\lambda} = \frac{6.63 \times 10^{-34} (3.0 \times 10^8)}{4(0.48 \times 10^{-9})} = 1.036 \times 10^{-16} \text{ J}$$

For formation of characteristic peaks, KE of electrons must be at least

$$E'_p = \frac{hc}{4\lambda} = \frac{6.63 \times 10^{-34} (3.0 \times 10^8)}{(0.64 \times 10^{-9})} = 3.108 \times 10^{-16} \text{ J}$$

Hence, the peaks are no longer observable.

30**C**